

Exercise 2

SQL Aggregate Functions & Operators.

1. SELECT DISTINCT department
FROM students;

students	department
IT	IT
HR	HR
Finance	Finance

2. SELECT department, AVG(age) AS avg_age
FROM students
GROUP BY department;

students	department	average_age
IT	IT	20.5
HR	HR	22
Finance	Finance	23

3. SELECT department, COUNT(*) AS student_count.
FROM student
GROUP BY department
HAVING COUNT(*) > 1;

students	department	student_count
IT	IT	2
HR	HR	2

4. SELECT *

FROM students

WHERE age BETWEEN 21 AND 23;

students			
student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. SELECT *

FROM students

WHERE department IN ('IT', 'HR')

AND age > 21;

students			
student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. SELECT department, sum(credits) AS total_credits

FROM courses

GROUP BY department

HAVING sum(credits) > 5;

courses	
department	total_credits
IT	11

```

8. SELECT *
   FROM courses
  WHERE credits < 4;

```

courses			
course-id	course-name	department	credits
101	SQL basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

```

9. SELECT course_id, course-name, credits
   FROM courses
  ORDER BY credits DESC
  LIMIT 3;

```

courses		
course-id	course-name	credits
102	Python	4
103	Data Science	4
101	SQL basics	3

```

9. SELECT MAX(grade) AS max-grade,
          MIN(grade) AS min-grade,
          AVG(grade) AS avg-grade
   FROM enrollments;

```

enrollments		
max-grade	min-grade	avg-grade
90	78	84.6

9. SELECT course-id, COUNT(*) AS enrollment-count
FROM enrollments
GROUP BY course-id;

enrollments

course-id	enrollments-count
101	1
102	1
103	1
104	1
105	1

11. SELECT department,
SUM(salary) AS total-salary,
SUM(Bonus) AS total-bonus
FROM salaries
GROUP BY department;

salaries

department	total-salary	total-bonus
HR	122 000	10 500
IT	109 000	7 500
Finance	70 000	6 000

12. SELECT department, AVG(salary) AS avg-salary
FROM salaries

GROUP BY department

HAVING AVG(salary) > 55000;

salaries	
department	avg-salary
IT	61000
Finance	70000

13. SELECT employee-id, name, salary, bonus, (salary + bonus) AS total-compens
FROM salaries

WHERE (salary + bonus) > 60000;

salaries				
employee-id	name	salary	bonus	total-compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

```

12. SELECT department, sum(budget) AS totalbudget,
      AVG(budget) AS avg-budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000;

```

project		
department	total_budget	avg-budget
IT	270000	135000
Finance	80000	80000

```

15. SELECT *
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department > 'Marketing';

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project			
project-id	project-name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000

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